clinical failure: random-effects meta-analysis (odds ratio)

Comparison: Intervention / Control

Methods

Odds ratios were pooled using a random-effects model with the DerSimonian–Laird estimator of between-study variance. Confidence intervals (95%) used the normal approximation. A continuity correction of 0.5 was applied to studies with any zero cell. 2 studies with no events in either arm were retained with this correction. Heterogeneity was assessed with Cochran's Q and I^2. Analyses were performed in RC MetaStudio 0.5.1. R packages loaded for this analysis: metafor 5.0.1, meta 8.5-0.

Results

19 studies were included (1097 participants in Intervention, 1080 in Control). The pooled odds ratio was 0.770 (95% CI [0.485, 1.222], z = -1.111, p = .267). The confidence interval includes 1, so these data do not show a clear difference between Intervention and Control. Heterogeneity: Q = 33.360, df = 18, p = .015. I^2 = 46.0%, which may represent moderate heterogeneity (Cochrane Handbook ranges). Between-study variance was τ^2 = 0.378. The 95% prediction interval was 0.212 to 2.799.

Table 1. Pooled effect

Effect	Estimate	95% CI	SE	z	p
Odds ratio	0.770	0.485 to 1.222	0.236	-1.111	.267

Table 2. Model fit

Statistic	Value
Studies (k)	19
Participants in Intervention	1097
Participants in Control	1080
Cochran’s Q	33.360
Heterogeneity degrees of freedom	18
Heterogeneity p	.015
I^2	46.0%
Between-study variance (τ^2)	0.378
95% prediction interval	0.212 to 2.799

Table 3. Study results

Study	Effect	95% CI	Weight
Gonzalez	0.571	0.170 to 1.916	7.3%
Prins	0.436	0.107 to 1.772	6.2%
Giamarellou	0.076	0.004 to 1.441	2.1%
Maller	1.401	0.667 to 2.945	10.7%
Sturm	0.189	0.009 to 4.080	2.0%
Marik	0.395	0.229 to 0.680	12.2%
Muijsken	1.611	0.491 to 5.289	7.5%
Vigano	2.878	0.115 to 71.821	1.8%
Hansen	0.140	0.028 to 0.704	5.3%
De Vries	1.973	0.475 to 8.188	6.1%
Mauracher	0.073	0.004 to 1.311	2.2%
Nordstrom	0.663	0.134 to 3.282	5.3%
Rozdzinski	1.018	0.464 to 2.236	10.3%
Ter Braak	1.356	0.501 to 3.666	8.7%
Tulkens	1	0.019 to 52.977	1.2%
Van der Auwera	0.187	0.009 to 4.062	2.0%
Klastersky	5.762	1.363 to 24.362	6.0%

Vanhaeverbeek	1.051	0.020 to 55.627	1.2%
Hollender	0.316	0.013 to 7.928	1.8%

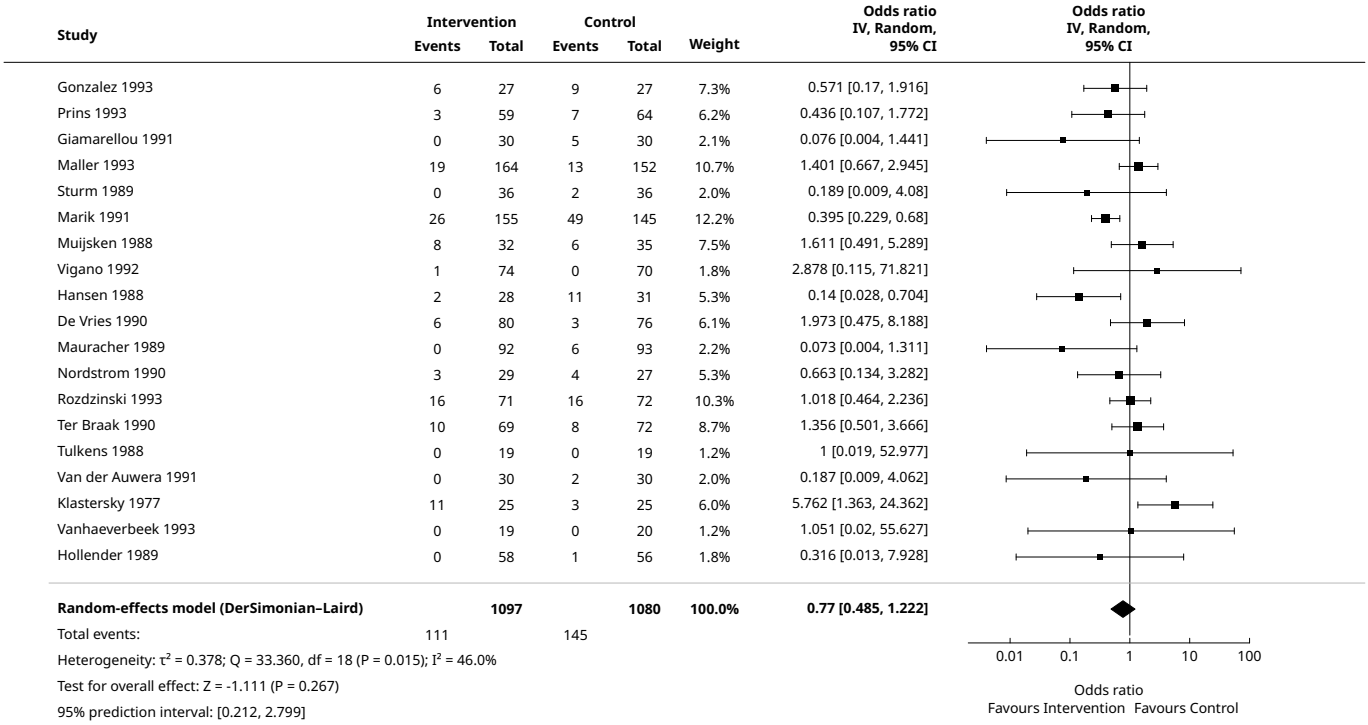

Figure 1. Forest plot

Supplementary details

Table 4. Full-precision results

Source	Results
Summary — Model Results	Estimate: 0.77 Lower bound (95% CI): 0.485 Upper bound (95% CI): 1.222 p-value: 0.267
Summary — Heterogeneity	Q: 33.360 df: 18 p-value: 0.015 I^2: 46.0% τ^2: 0.378
Summary — Test for overall effect	Z: -1.111 p-value: 0.267
Summary — Prediction interval	Lower bound (95% PI): 0.212 Upper bound (95% PI): 2.799
Summary — Calculation scale (log)	Estimate: -0.262 Lower bound (95% CI): -0.724 Upper bound (95% CI): 0.200 SE (log scale): 0.236
Weights — Study weights	Study: Gonzalez Effect: 0.571 Lower bound: 0.17 Upper bound: 1.916 Weight (%): 7.3%
Weights — Study weights	Study: Prins Effect: 0.436 Lower bound: 0.107 Upper bound: 1.772 Weight (%): 6.2%
Weights — Study weights	Study: Giamarellou Effect: 0.076 Lower bound: 0.004 Upper bound: 1.441 Weight (%): 2.1%
Weights — Study weights	Study: Maller Effect: 1.401 Lower bound: 0.667 Upper bound: 2.945 Weight (%): 10.7%
Weights — Study weights	Study: Sturm Effect: 0.189 Lower bound: 0.009 Upper bound: 4.08 Weight (%): 2.0%
Weights — Study weights	Study: Marik Effect: 0.395 Lower bound: 0.229 Upper bound: 0.68 Weight (%): 12.2%
Weights — Study weights	Study: Muijsken Effect: 1.611 Lower bound: 0.491 Upper bound: 5.289 Weight (%): 7.5%
Weights — Study weights	Study: Vigano Effect: 2.878 Lower bound: 0.115 Upper bound: 71.821 Weight (%): 1.8%
Weights — Study weights	Study: Hansen Effect: 0.14 Lower bound: 0.028 Upper bound: 0.704 Weight (%): 5.3%

Weights — Study weights	Study: De Vries Effect: 1.973 Lower bound: 0.475 Upper bound: 8.188 Weight (%): 6.1%
Weights — Study weights	Study: Mauracher Effect: 0.073 Lower bound: 0.004 Upper bound: 1.311 Weight (%): 2.2%
Weights — Study weights	Study: Nordstrom Effect: 0.663 Lower bound: 0.134 Upper bound: 3.282 Weight (%): 5.3%
Weights — Study weights	Study: Rozdzinski Effect: 1.018 Lower bound: 0.464 Upper bound: 2.236 Weight (%): 10.3%
Weights — Study weights	Study: Ter Braak Effect: 1.356 Lower bound: 0.501 Upper bound: 3.666 Weight (%): 8.7%
Weights — Study weights	Study: Tulkens Effect: 1 Lower bound: 0.019 Upper bound: 52.977 Weight (%): 1.2%
Weights — Study weights	Study: Van der Auwera Effect: 0.187 Lower bound: 0.009 Upper bound: 4.062 Weight (%): 2.0%
Weights — Study weights	Study: Klastersky Effect: 5.762 Lower bound: 1.363 Upper bound: 24.362 Weight (%): 6.0%
Weights — Study weights	Study: Vanhaeverbeek Effect: 1.051 Lower bound: 0.02 Upper bound: 55.627 Weight (%): 1.2%
Weights — Study weights	Study: Hollender Effect: 0.316 Lower bound: 0.013 Upper bound: 7.928 Weight (%): 1.8%

Pooled Odds ratio: 0.77 (95% CI 0.485 to 1.222).

95% prediction interval: the underlying effect in a new study; this is not the confidence interval of the pooled estimate.

References

1. Random-effects meta-analysis: DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. Controlled Clinical Trials, 7(3), 177-188. doi:10.1016/0197-2456(86)90046-2.

2. Random-effects meta-analysis: Hedges, L. V. (1983). A random effects model for effect sizes. Psychological Bulletin, 93(2), 388-395. doi:10.1037/0033-2909.93.2.388.

3. Implementation reference: Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. doi:10.18637/jss.v036.i03.